

Bariş Salih Babacan

Software Engineer & Systems Architect

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PROFESSIONAL SUMMARY

Innovative Software Engineer and Systems Architect specializing in autonomous systems, local-first (P2P) architectures, and high-performance web/mobile applications. Proven track record of owning the full software development lifecycle—from conceptualizing system architectures to deploying production-ready platforms. Adept at integrating complex hardware-software systems and migrating legacy codebases to modern, scalable ecosystems.

CORE COMPETENCIES

- **Languages:** TypeScript, JavaScript, Rust, C++, Python, PHP, SQL
- **Frontend & Mobile:** React, Next.js, SwiftUI, Vite, D3.js, TailwindCSS
- **Backend & Systems:** Node.js, Firebase (Firestore, Functions), SQLite, NoSQL, Tauri, CRDTs (P2P Sync)
- **Tools & DevOps:** Git, Playwright (Stealth), Docker, WireShark, Postman, Figma

PROFESSIONAL EXPERIENCE

CTO & Lead Developer | *Garage.ist* | Present

- Orchestrated the full development lifecycle and App Store launches for flagship applications like **Sortify** and **+TV**.
- Architected a proprietary network-based TV control protocol and developed the core system from the ground up.
- Led a cross-functional technical strategy utilizing SwiftUI for native iOS interfaces and Node.js/TypeScript for robust backend APIs.

IT & Systems Integration (Intern) | *ENLOG - Istanbul* | Sep 2023 – Jun 2024

- Handled technical support, network operations, and workflow optimization within the IT infrastructure.

SELECTED PROJECTS & ARCHITECTURES

- **Osmos (P2P / Local-First)** | *Rust, SwiftUI, CRDTs* Architected a peer-to-peer version control and sync system for the Apple ecosystem. Built with a high-performance **Rust core** and a native **SwiftUI layer**, completely eliminating cloud dependency for secure, offline-first data sovereignty.
- **Mythos IDE** | *Tauri, TypeScript, AI/MCP* Created a specialized, local-first development environment for narrative designers. Engineered scalable storytelling hierarchies and integrated advanced AI orchestration directly into the IDE.
- **Gayrimenkul Dünyası** | *Next.js, Playwright Stealth, Firestore* Built a high-performance automated data aggregator. Implemented a complex scraping pipeline using Playwright Stealth alongside a fallback Google News XML parser, synchronized with Cloud Firestore using daily transaction budgets.
- **Q-PASS Access Control System** | *C++, Python, PHP, MySQL, Arduino* Engineered a live, end-to-end hardware-software security system for a public institution. Bridged C++ microcontroller firmware with a Python/PHP web backend via serial communication, implementing strict RBAC and 21 security rules.

- **IT-RMS (Resource Management System)** | *Node.js, NoSQL, PHP* Modernized a monolithic PHP application by migrating it to a scalable Node.js/NoSQL architecture. Implemented automated barcode generation pipelines and multi-branch access control logic.

EDUCATION

Istanbul University (AUZEF) | *Computer Programming (A.S.)* | 2024 – Present

Haydarpaşa Vocational & Technical High School | *Network Management & Cyber Security* | 2020 – 2024